COURSE DESIGN

Key considerations for planning and laying out the run course

CLOSED vs. OPEN COURSE

Closed courses are fully or partially restricted to outside traffic — requiring detailed traffic control plans. Open courses require delineation equipment and public safety personnel to safely separate athlete and vehicle lanes. Park paths may still need robust signage and marshaling despite no vehicle traffic.

ROAD WIDTHS

Road width must accommodate the peak number of athletes on course at any time, with sufficient room for runners to safely execute passing maneuvers without interference.

TERRAIN

Surfaces can vary: pavement, grass, gravel, trail, sand. Always communicate terrain types to athletes in advance so they can prepare appropriate footwear. Avoid terrain changes near sharp turns or steep descents to minimize slipping hazards.

DID YOU KNOW?

Per USAT rules, an athlete may run barefoot or without any footwear on any part of the run course (except youth events).

Pro Tip — Para / Adaptive Inclusivity

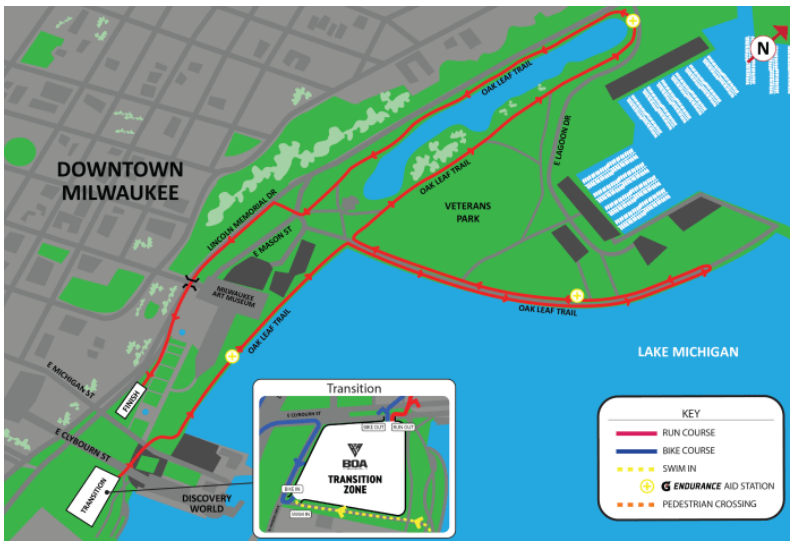
Unpaved terrain, curbs, bumps, and stairs can make the race less accessible for para/adaptive triathletes. Have conversations with prospective para/adaptive athletes before the event so they can make an informed decision about participating.

MULTI-DISTANCE COURSE

Running concurrent sprint and Olympic distances? Clearly mark all split points where courses diverge. Use distinct color schemes per distance on signage to prevent confusion and ensure every athlete follows the correct route.

MULTI-LOOP COURSE

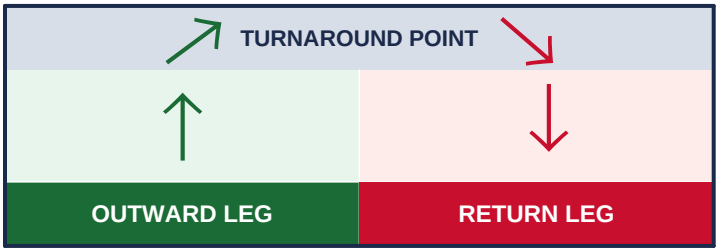
Multi-loop designs reduce road closures and resource needs while boosting spectator access. However, they increase congestion risk. Balance the spectator benefit against the course capacity — athletes must always be able to run freely. See the Bike Course Logistics Resource Guide for additional guidance.



OUT-AND-BACK SECTIONS

An **out-and-back** section goes *"out"* from a start to a turnaround point, then returns *"back"* along the same path.

- Delineation equipment (cones, barricades, delineators) is essential to clearly separate athletes traveling in opposite directions.
- Timing mats at the turnaround are critical to prevent course cutting and enhance the spectator experience with live updates.



✓ PROS	✗ CONS
✓ Athletes gauge race position and build camaraderie passing competitors.	✗ Repetitive nature can feel less engaging than varied course designs.
✓ Simple layout — easy for participants, staff, and volunteers to follow.	✗ Congestion risk when athletes travel in both directions simultaneously.
✓ Maximum access for spectators, media, and broadcast coverage.	✗ Easier for athletes to cheat by cutting the course short.*

TIMING MAT STRATEGY

**For multi-loop and out-and-back courses, positioning timing mats at key checkpoints (especially the furthest turnaround point) is essential to maintain course integrity and prevent cutting. Additional mats enhance the spectator experience by triggering live updates for friends and family via event apps.*

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